Northeastern University
Khoury College of Computer Sciences

CS5800: Algorithms
Syllabus

General Information

Instructor	Jonathan Mwaura
Office	Online (Via Teams – See Canvas)
Email	j.mwaura@northeastern.edu
Prof Office Hours	Tue (4.30pm – 6pm) and Wed (3pm – 5pm)
Class Time/ Location	Roux 136 & Roux 137
TA/Grader	Rithwik Katuri (katuri.r@northeastern.edu)
	Tuoxin Li (li.tuox@northeastern.edu)
	Armaan Sait (sait.ar@northeastern.edu)
	Dilshath Shaik (shaik.di@northeastern.edu)
	Mian Wang (wang.mian1@northeastern.edu)

Table 1: General Information

Contact the Instructor

- Office hours: the best way to contact the professor is to visit the professor's office hours.
- Ms Teams Classes: – Office hours and Q &A: Please click on "Microsoft Teams Canvas to access the Teams channel. You do not need to ask to join.
  Please post questions related to assignments and class on the Teams discussion channels for each module whenever possible, so others can benefit too. Read the rules further below.
- Email: If you need to send an email to discuss private issues (e.g., your grades), please keep the message brief and put "[CS5800]" in the email title. Otherwise, your email may not be read and replied in a timely manner. Please don't expect the emails will be read as soon as you send them out (leave at least 48 hours before re-emailing).

Required Textbook

Zybooks

1. Sign in or create an account at learn.zybooks.com
2. Enter zyBook code: NORTHEASTERNCS5800MwauraFall2023
3. Subscribe

Supplementary Textbook

- Algorithm Design by Jon Kleinberg and Eva Tardos (This is the book used by Prof Ravi in the pre-recorded videos). It is a very well written book.
- *Introduction to Algorithms* by T.H. Cormen, C.E. Leiserson, R.L. Rivest, MIT Press, 3rd edition, 2009. ISBN 978-0-262-03384-8.

Course Description

This is a graduate-level course that emphasizes the design and analysis of efficient algorithms used in software engineering. The course focuses on the algorithmic design paradigms and techniques for analyzing the correctness, time, and space complexity of algorithms. The topics covered include elementary techniques for analysis (Asymptotic analysis, recursion equations, estimation methods etc), Analysis of Sorting Algorithms (Heap Sort, Quick Sort, Selection, Bucket/Radix etc), Analysis of various Abstract Data structures (Stacks, Queues, Linked Lists, Trees including Binary Search Trees and self-balancing trees) and Hash maps. Further, different strategies to solving problems including divide and conquer, greedy technique and dynamic programming will be explored. Additionally, Analysis of different Graph Algorithms and graph-based problems (Breadth First Search, Depth First Search and Topological Sort, Shortest path problems, Network flow problems etc.). Finally, a discussion revolving around NP Completeness and reducibility will be covered.

Course Category

This is a required course for students pursuing an MSC in Computer Science. It builds on the introduction to data structures and algorithms that students receive in both Data Structures and Algorithms (CS 5004/ CS3000) class and the discrete structures class (CS1800/CS5002) or their equivalents.

Course Core Learning Outcomes

- This course meets the Essential Learning Outcome of Exploring Creative Expression and Innovation as defined under the NU's Core Curriculum requirements. As such, it will reinforce the students' ability to identify, analyze, interpret, and evaluate arguments, data, evidence, problems, and conclusions as part of formulating an opinion or conclusion. Then use that information to design, evaluate and implement a strategy to achieve a desired outcome.
- This course meets the Essential Learning Outcome of Conducting Formal and Quantitative Reasoning as defined under the NU's Core Curriculum requirements. As such, the course will strengthen the students' competency and comfort in working with numerical data.
- This course meets the Essential Learning Outcome of Analyzing and Using Data as defined under the NU's Core Curriculum requirements. As such, it provides students with the opportunity to practice the skills needed to successfully locate, evaluate, and use data, a fundamental ability for scholars and citizens operating in a complex, global information landscape.

Course Prerequisites

The course assumes that students have taken introductory CS mathematics courses (Discrete, Probability and Statistics and Calculus). In addition, students are expected to have taken a Data Structures and Algorithms class (CS5004 /CS3000) or an equivalent course.

Additional Course Information

Class Website – on Canvas. All class materials (including lecture notes, supplemental materials, etc.). Assignments, however, are posted on Canvas and Gradescope. **The course modules unlock every Monday at 00.00 hours (i.e., at around 11.59 on a Sunday)**

Class Discussion Website – (via MS Teams Classes)-Teams can be accessed directly from Canvas. Download the app for better experience on your computer or mobile device.

Teams can be used to clarify your question about assignments and discuss general questions. In

addition the Prof uses Ms Teams to enhance and foster continuous learning outside the classroom. When possible, students should ask related questions through this portal instead of sending individual emails to the instructor. Please do NOT post your programming code or solutions there. If you have a specific question about your solution, please ask help from the instructor, TA, or approved tutors in their offices.

Tentative class schedule

Week	Week Dates	Discussion Topic	Associated Work (Due on Wednesdays at 11.30pm)	
1 & 2	Sept 7th and 14th	Introduction to Algorithms and Algorithm Analysis including Growth of Functions	Problem Set 1	Quiz 1
3	Sept 21st	Divide and Conquer 1	Problem Set 2	Quiz 2
4	Sept 28th	Divide and Conquer 2	Problem Set 3	Quiz 3
5	Oct 5th	Data Structures	Problem Set 4	Quiz 4
6	Oct 12th	Advanced Data Structures	Problem Set 5	Quiz 5
7	Oct 19th	Greedy Algorithms 1 & 2	Problem Set 6	Quiz 6
8	Oct 26th	Revision Week	Synthesis 1	
9	Nov 2nd	Dynamic Programming 1	Problem Set 7	Quiz 7
10	Nov 9th	Dynamic Programming 2	Problem Set 8	Quiz 8
11	Nov 16th	Network Flow Problems	Problem Set 9	Quiz 9
12	Nov 23rd	***Thanksgiving Break ***		
13	Nov 30th	NP Completeness	Problem Set 10	Quiz 10
14	Dec 7th	Approximation Algorithms		
15	Dec 14th	Final project presentations	Final Project Report / Synthesis 2	

Table 2:Tentative class schedule

Methodology

Teaching methods:

This course will make use of lectures (videos), examples, and question and answer sessions. Class discussions conducted via an in-class medium (either in small group or with whole class), in addition to active reading, experimenting with sample programs, and problem solving, will be used during the semester. Independent homework assignments are used to evaluate the learning outcomes.

While this is not a programming class, an extensive knowledge of programming is required. Students are expected to read code, implement code and analyze code. Students are also encouraged to implement their algorithms throughout the learning period. During the in-class discussion, the professor and TA's will use pseudocode and/or source code to explain different data structures etc.

Assessment:

There will be regular weekly exercises, problem sets, and quizzes as described below.

- Readings – Assigned Weekly (10%) – Readings will help strengthen your understanding of the material that we are covering in class.

- **10 timed online Quizzes (10%)**. These are timed released either via Canvas or Grade scope. Note when you start the quiz, you will have up to 240 minutes to finish it after which it will lock. Students are required to allocate 45-60 minutes to work on the quiz.
- **10 Problem Sets (45%)** – These are weekly homework assignments that are to be carried out individual. These will involve programming and written analysis. Pair programming is encouraged for programming pieces. Note, lowest scored problem set will be dropped.
- **2 Course Syntheses (20%)** consist of short answer questions, as well as several multi-part problems connecting different areas of the course, allowing you to synthesize what you have learned. Think of the Course Synthesis as a week-long individual take-home exam where you may consult your class notes but not your classmates or any online resources other than the ones that are explicitly permitted by the course instructor.
- **4 SAIL Reflections (5%)** are your personal reflections on your journey of self-authored integrated learning (SAIL) in this course. You will reflect on your growth across five learning dimensions: Intellectual Agility, Global Mindset, Social Consciousness and Commitment, Professional and Personal Effectiveness, and Well-Being. For more details, check out https://sail.northeastern.edu/about/.
- **1 Final Project (10%)** work occurs during the last two weeks of the course. You are required to work within a group with other learners to create a project. Each group will select any topic relating to the design and/or analysis of Algorithms. Your group will submit a project proposal, a written report, and send in videos where you are presenting your work.

Final Grades:

A	93.00 – 100.00 %
A -	90.00 – 92.99%
B+	86.00 – 89.99%
B	83.00 – 85.99%
B-	80.00 – 82. 9%
C+	76.00 – 79.99%
C	73.00 – 75.99%
C-	70.00 – 72.99%
F	Less that 70%

Table 3: Final grading

Re-grading Assignments:

Requests for re-grading may be made up to one week after the assignment is returned. Please contact the TA first if you have any questions about the grading. If you still have questions, the request must be made in person with the instructor or be submitted in writing and include a short paragraph outlining the rationale for the re-grade. Acceptable requests include correcting errors in calculating a score, marking a correct answer incorrect, etc.

Course Policies

Accommodations:

The goal is for every student to succeed in this course. If you require any accommodations (e.g. child care during class hours, extra time to complete assignments, support for a disability), let me know immediately so that we can work out appropriate arrangements. Speak to me at the end of class or contact me by email, and we will set up a time to meet

during the first week of the course. I look forward to learning how I can be of service to you.

Attendance:

This is a traditional set up class. Students are expected to attend the class in person in the scheduled time and place. However, all class materials will be released weekly via online mechanism, and students are expected to complete the work within the given timelines.

Course readings:

Required readings will be announced and scheduled as assignments on the canvas platform. All students are required to purchase the required textbook. It is a student's objective to complete all required reading and the ensuing assignments. Note that assignments related to the readings will not be extended.

Assignments:

Assignments will be distributed via the Canvas and Gradescope platforms. Students are responsible for managing due dates and understanding submission procedures to turn in assignments. All assignments are due at 11.30pm on Tuesdays.

Late work:

Late assignments will not be accepted without PRIOR approval. Students must consult the instructor **at least two days prior to the scheduled due date to make alternative arrangements;** however, the instructor is under no obligation to grant any such request.

There are two different deadlines. **Due date (early deadline or just the deadline)** is the due date when the assignments is due. The **late due date (or Late deadline)** is the last date when you can turn in the work for credit. Late due date is typically 48 hours after the due date. You won't be penalized for turning the assignment using the late deadline. However, any work past the late deadline will receive zero (0) credit.

When a student is requesting for additional time to take an assignment; the professor will be basing this on the Due Date. Note that 1 minute late is considered within the 2 days (48 hours) late due date allowance. Internet or computer issues are NOT valid excuses of late submission.

Note that this late due date only applies to Problem Sets, Programming Projects, and Quizzes. The remaining assessments (Synthesis, Personal Reflections and Surveys, Group Seminar, Final Project) must be submitted on time; failure to do so will result in an automatic zero.

Note that, assignments MUST be submitted through Gradescope. Professor does not accept emailed assignments or assignments submitted via canvas unless specifically asked to.

Exams:

The course will NOT have formal timed exams:

Academic integrity:

The practice of good ethical behavior is essential for maintaining good order in the classroom, providing an enriching learning experience for students, and as training as a practicing

computing professional upon graduation. This practice is manifested in the University's Academic Integrity policy. Students are expected to strictly avoid academic dishonesty and adhere to the Academic Integrity policy as outlined in the course catalog. Violations will be dealt with as outlined therein.

As a general rule, all work submitted for grading must be the student's OWN work.

In regard to homework, students may discuss the problems (what is being asked for), appropriate material from class lectures, the textbook or acceptable other sources. Students, however, may not share answers or the specifics of how to answer the question.

Use of material from previous classes, solution manuals, material from the Internet or other sources (e.g., parents, siblings, friends, etc.) that directly bears on the answer is strictly prohibited. Please cite the references if you use any help.

In addition, you are not allowed to post course materials and solutions to problem sets assigned in this class in public places (e.g. Github, courseHero) without the instructor's permission. Solutions include your own solutions as well as solutions that may be provided by the instructor. The University policy on academic integrity states that assisting students in their own acts of academic dishonesty is itself a violation of academic integrity. Doing so will be considered an act of academic dishonesty and you will receive a grade of 0 for the actual assignment or a grade of F for the course.

At the discretion of the instructor, students may be asked to sign a statement that they have abided by the University's Academic Integrity policy and its application to this class. This statement may appear on homework, or tests.

When in doubt, consult the course professor before doing something that may result in violation of the University's Academic Integrity policy (http://www.northeastern.edu/osccr/academic-integrity)

The sanction for the first violation of the Academic Integrity policy or plagiarism policy will result in a minimum failing grade on the relevant assignment and the violation will be reported to the student's department chair. Once the final decision has been rendered and any or all appeals exhausted by any parties involved, the instructor or appropriate parties will carry out the recommended sanction.

Title IX Policy

Title IX of the USA Education Amendments of 1972 protects individuals from sex or gender-based discrimination, including discrimination based on gender-identity, in educational programs and activities that receive federal financial assistance. Though our campus is located in Canada, all Northeastern University campuses follow the Title IX Policy.

Northeastern's Title IX Policy prohibits Prohibited Offenses, which are defined as sexual harassment, sexual assault, relationship or domestic violence, and stalking. The Title IX Policy applies to the entire community, including male, female, transgender students, faculty and staff.

If you or someone you know has been a survivor of a Prohibited Offense, confidential support and guidance can be found through University Health and Counseling Services staff and the Center for Spiritual Dialogue and Service clergy members. By law, those employees are not

required to report allegations of sex or gender-based discrimination to the University.

Alleged violations can be reported non-confidentially to the Title IX Coordinator within The Office for Gender Equity and Compliance at: titleix@northeastern.edu and/or through NUPD Emergency 617.373.3333; Non-Emergency 617.373.2121. Reporting Prohibited Offenses to NUPD does NOT commit the victim/affected party to future legal action.

Faculty members are considered "responsible employees" at Northeastern University, meaning they are required to report all allegations of sex or gender-based discrimination to the Title IX Coordinator.

In case of an emergency, please call 911. Please visit http://www.northeastern.edu/titleix for a complete list of reporting options and resources, both on-campus and off-campus.

Feedback

Your opinions are very important to me. All students are strongly encouraged to use the Teacher Rating and Course Evaluation (TRACE) system, at https://www.northeastern.edu/trace/, to complete your course evaluations. A reminder about TRACE should arrive via email about two weeks before the end of the course.

In addition, I will be asking for your feedback at least once a month to see how the course is going. I will also do week reflection via Teams to gauge how the learning is going on. However, if you have concerns about the course, do not wait until you are asked. Please schedule a meeting with me, and we will discuss your concerns then.

Thank you for taking this course and entrusting me to shape your education here at Northeastern.